Inorganic / Inorganic Materials Chemistry

Cumulative Exam

10-15-21

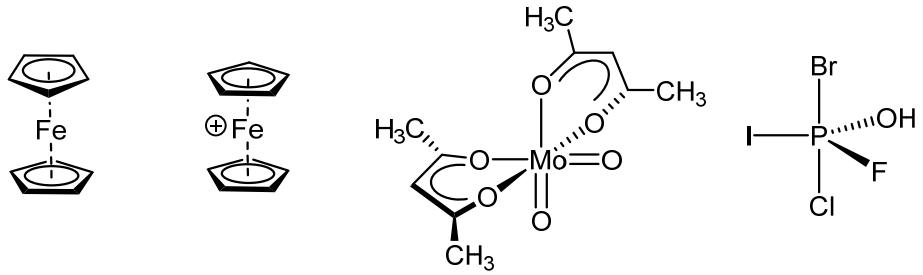
Instructions: Answer any five of the following questions. Only your first five answers will be graded.

Homogeneous inorganic chemistry.

1. (20 pts) Describe the experimental evidence Alfred Werner used to demonstrate that the complex ion theory of inorganic coordination compounds was valid and the "valence chain" theory of inorganic compounds was not.

2. (20 pts) Assign point groups to the following molecules

A B C D



3. (20 pts) Assign oxidation states, d electron counts and the total number of valence electrons in complexes A-D of question 2.

4. (20 pts) Describe the ^{31}P NMR spectrum and the ^{19}F NMR spectrum of the hexafluorophosphate anion, PF_6^- (reminder, both ^{31}P and ^{19}F are over 99% abundant and for both nuclei I=1/2)

5. Draw a complete molecular orbital diagram for F_2 include labels for each of the MOs that summarizes their symmetry (sigma, pi, delta, g,u etc.)

Heterogeneous inorganic chemistry.

6. Alkaline batteries are ubiquitously available and utilize the reaction of MnO_2 to Mn_2O_3 as well as zinc metal to form a galvanic cell. Provide a balanced redox equation for an alkaline battery.

7. A simple galvanic cell can be assembled using a copper electrode submerged in a 1M solution containing Cu^{2+}, and a zinc electrode submerged in a 1M solution containing Zn^{2+}. The two electrodes are connected with a metallic wire and the two solutions are in ionic contact via a salt bridge or an appropriate permeable membrane.

a) Which electrode is the anode and which electrode is the cathode?
b) In an ideal system what is the cell potential?
c) Describe the flow of electrons and ions in this system.
d) What would happen if the Cu^{2+} solution was replaced with 1M solution of sodium nitrate?

8. Mixed oxides that crystalize in a spinel structure are popular materials in some battery electrodes. Their chemical formulas are summarized as AB_2O_4 where A and B can be a variety of cations. The "normal" spinel structure can be described as a face centered cubic (*fcc*) array of O^{2-} with A occupying a fraction of the tetrahedral holes and B occupying a fraction of the octahedral holes. How many Total tetrahedral and octahedral holes, respectively, exist per O^{2-} ion in an *fcc* array?

9. Assuming a reversible outer sphere one electron transfer between a generic species "A" and an inert electrode like boron doped diamond, describe how the cyclic voltammogram of species A would change if the heterogeneous electron transfer rate constant were to change from 1 cm/s to 0.001 cm/s to 0.000001 cm/s. This answer does not have to be quantitative but should give in impression of what you expect. Please include an approximate sketch of the expected voltammograms. (Hint, focus your attention on the diffusion limited peak potential of the reduction and oxidation processes)

10. Define the exchange current density for a heterogeneous electrocatalyst. What can be inferred when a heterogeneous catalyst or material has a much larger exchange current density compared to another?

Standard Reduction Potentials in Aqueous Solutions at 25 °C

Oxidizing Agent				Reducing Agent	Reduction Potential (V)
F_2	+	$2e^-$	→	$2F^-$	2.87
H_2O_2	+	$2H^+ + 2e^-$	→	$2H_2O$	1.78
MnO_4^-	+	$8H^+ + 5e^-$	→	$Mn^{2+} + 4H_2O$	1.51
Au^{3+}	+	$3e^-$	→	Au	1.50
Cl_2	+	$2e^-$	→	$2Cl^-$	1.36
O_2	+	$4H^+ + 4e^-$	→	$2H_2O$	1.23
$Cr_2O_7^{2-}$	+	$14H^+ + 6e^-$	→	$2Cr^{3+} + 7H_2O$	1.23
Br_2	+	$2e^-$	→	$2Br^-$	1.07
NO_3^-	+	$4H^+ + 3e^-$	→	$NO + 2H_2O$	0.96
Ag^+	+	e^-	→	Ag	0.80
I_2	+	$2e^-$	→	$2I^-$	0.54
Cu^+	+	e^-	→	Cu	0.52
O_2	+	$2H_2O + 4e^-$	→	$4OH^-$	0.40
Cu^{2+}	+	$2e^-$	→	Cu	0.34
$2H_3O^+$	+	$2e^-$	→	$H_2 + 2H_2O$	0.00
Pb^{2+}	+	$2e^-$	→	Pb	−0.13
Sn^{2+}	+	$2e^-$	→	Sn	−0.14
Ni^{2+}	+	$2e^-$	→	Ni	−0.26
Fe^{2+}	+	$2e^-$	→	Fe	−0.45
Cr^{3+}	+	$3e^-$	→	Cr	−0.74
Zn^{2+}	+	$2e^-$	→	Zn	−0.76
$2H_2O$	+	$2e^-$	→	$H_2 + 2OH^-$	−0.83
Mn^{2+}	+	$2e^-$	→	Mn	−1.19
Al^{3+}	+	$3e^-$	→	Al	−1.66
Mg^{2+}	+	$2e^-$	→	Mg	−2.37
Na^+	+	e^-	→	Na	−2.71
Ca^{2+}	+	$2e^-$	→	Ca	−2.87
Ba^{2+}	+	$2e^-$	→	Ba	−2.91
K^+	+	e^-	→	K	−2.93
Li^+	+	e^-	→	Li	−3.04

Increasing Strength of Oxidizing Agent

Increasing Strength of Reducing Agent

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